Ghosh's Autonomous Ordinary Differential Equation

Closed-Form Solutions, Stability Analysis, and Computational Methods

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Abstract

We present a complete analytical framework for the autonomous ordinary differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = a\tanh(y) + b\tanh(\tau - y) + c,$$

henceforth referred to as *Ghosh's Equation*. This first-order nonlinear ODE arises naturally in thermal diffusion processes, neural network dynamics, and population models with saturation effects. We establish the existence of closed-form solutions via rational function integration, characterize equilibrium points and their stability, develop efficient numerical algorithms, and provide a complete classification of solution behaviors based on parameter regimes. The interplay between the two hyperbolic tangent terms with shifted arguments produces rich dynamical behavior, including bistability, heteroclinic connections, and parameter-dependent bifurcations.

The paper ends with "The End"

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1 Introduction

The study of autonomous ordinary differential equations constitutes a foundational pillar of dynamical systems theory. Among the myriad forms such equations can take, those involving hyperbolic functions occupy a distinguished position due to their natural emergence in physical systems exhibiting saturation phenomena.

Definition 1.1 (Ghosh's Autonomous ODE)**.** Let $a, b, c \in \mathbb{R}$ be constants and $\tau \in \mathbb{R}$ be a shift parameter. **Ghosh's Equation** is the first-order autonomous ordinary differential equation

$$\boxed{\frac{\mathrm{d}y}{\mathrm{d}x} = a\tanh(y) + b\tanh(\tau - y) + c} \tag{1}$$

where $y\colon I \subseteq \mathbb{R} \to \mathbb{R}$ is the unknown function and x denotes the independent variable.

The equation derives its structure from the superposition of two sigmoidal responses: one centered at $y = 0$ and another centered at $y = \tau$. This configuration models systems in which two competing saturation mechanisms operate at distinct reference points. Applications include thermal systems with boundary layers, neural activation dynamics, and ecological models with carrying capacities at multiple scales.

Right-Hand Side of Ghosh's Equation

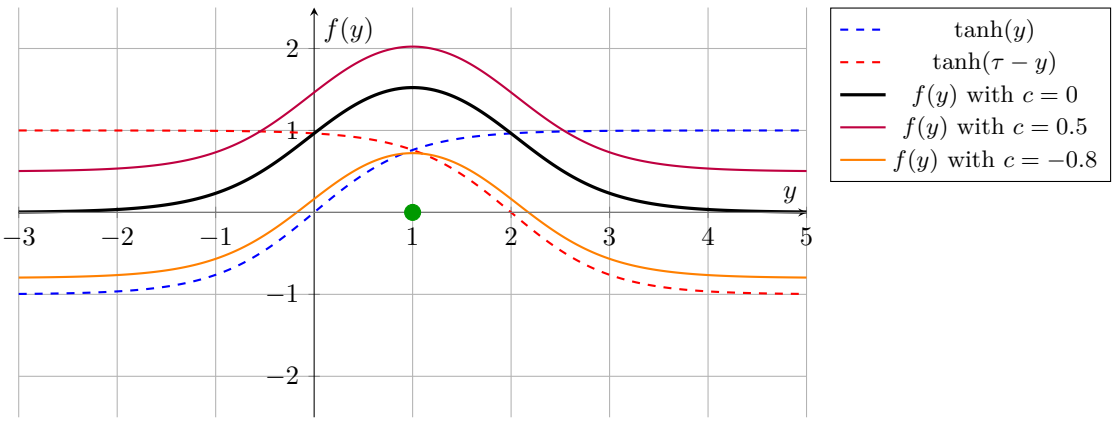

Figure 1: The function $f(y) = a\tanh(y) + b\tanh(\tau - y) + c$ plotted for $a = b = 1$, $\tau = 2$, and various values of c.

subcaption*Equilibria occur at intersections with the horizontal axis.

2 Mathematical Preliminaries

2.1 Hyperbolic Function Identities

The following identities play a central role in our analysis.

Lemma 2.1 (Fundamental Hyperbolic Identities)**.** *For all $y, \tau \in \mathbb{R}$, the following hold:*

$$\tanh(y) = \frac{e^{2y} - 1}{e^{2y} + 1} = \frac{\sinh(y)}{\cosh(y)}, \tag{2}$$

$$\tanh(y) + \tanh(\tau - y) = \frac{\sinh(\tau)}{\cosh(y)\cosh(\tau - y)}, \tag{3}$$

$$\tanh(y) - \tanh(\tau - y) = \frac{\sinh(2y - \tau)}{\cosh(y)\cosh(\tau - y)}, \tag{4}$$

$$\cosh(y)\cosh(\tau - y) = \frac{1}{2}\big[\cosh(\tau) + \cosh(2y - \tau)\big]. \tag{5}$$

Proof. Identity (2) follows directly from the definitions $\sinh(y) = (e^y - e^{-y})/2$ and $\cosh(y) = (e^y + e^{-y})/2$.

For (3), we compute

$$\begin{aligned}\tanh(y) + \tanh(\tau - y) &= \frac{\sinh(y)}{\cosh(y)} + \frac{\sinh(\tau - y)}{\cosh(\tau - y)} \\ &= \frac{\sinh(y)\cosh(\tau - y) + \sinh(\tau - y)\cosh(y)}{\cosh(y)\cosh(\tau - y)} \\ &= \frac{\sinh\big((y) + (\tau - y)\big)}{\cosh(y)\cosh(\tau - y)} = \frac{\sinh(\tau)}{\cosh(y)\cosh(\tau - y)},\end{aligned}$$

where we applied the addition formula $\sinh(A + B) = \sinh A \cosh B + \cosh A \sinh B$.

Identity (4) follows analogously using $\sinh(A - B) = \sinh A \cosh B - \cosh A \sinh B$ with $A = y$ and $B = \tau - y$.

For (5), we invoke the product-to-sum formula

$$\cosh A \cosh B = \frac{1}{2}\big[\cosh(A + B) + \cosh(A - B)\big]$$

with $A = y$ and $B = \tau - y$, yielding $\cosh(A + B) = \cosh(\tau)$ and $\cosh(A - B) = \cosh(2y - \tau)$. □

2.2 Existence and Uniqueness of Solutions

Theorem 2.2 (Global Existence and Uniqueness)**.** *For any initial condition $y(x_0) = y_0$ with $x_0, y_0 \in \mathbb{R}$, Ghosh's Equation* (1) *admits a unique solution $y\colon \mathbb{R} \to \mathbb{R}$ defined for all $x \in \mathbb{R}$.*

Proof. Define $f(y) := a\tanh(y) + b\tanh(\tau - y) + c$. Since tanh belongs to $C^\infty(\mathbb{R})$, so does f. The derivative is

$$f'(y) = a\,\mathrm{sech}^2(y) - b\,\mathrm{sech}^2(\tau - y),$$

which is continuous and bounded on $\mathbb{R}$ because $0 < \mathrm{sech}^2(\cdot) \le 1$. Consequently, f is globally Lipschitz continuous with constant $L = |a| + |b|$:

$$|f(y_1) - f(y_2)| \le L|y_1 - y_2| \quad \text{for all } y_1, y_2 \in \mathbb{R}.$$

By the Picard–Lindelöf theorem, a unique local solution exists through any initial point. Furthermore, since $|f(y)| \le |a| + |b| + |c|$ for all y, solutions cannot escape to infinity in finite time. This precludes finite-time blow-up, thereby guaranteeing global existence on $\mathbb{R}$. □

3 Closed-Form Solutions

3.1 The Fundamental Integration Problem

Being autonomous and first-order, Ghosh's Equation is separable. The general solution satisfies the implicit relation

$$x - x_0 = \int_{y_0}^{y} \frac{\mathrm{d}s}{a\tanh(s) + b\tanh(\tau - s) + c} =: \mathcal{I}(y; y_0). \tag{6}$$

The central question is whether this integral admits a closed-form antiderivative.

Theorem 3.1 (Closed-Form Integrability)**.** *The integral $\mathcal{I}(y; y_0)$ in* (6) *can be expressed in closed form using elementary functions—specifically, logarithms and inverse trigonometric functions—for all parameter values $a, b, c, \tau \in \mathbb{R}$ such that the denominator is not identically zero.*

Proof. We employ the exponential substitution $u = e^{2y}$, so that $\mathrm{d}y = \mathrm{d}u/(2u)$. Each hyperbolic tangent transforms as follows:

$$\begin{aligned}\tanh(y) &= \frac{u - 1}{u + 1}, \\ \tanh(\tau - y) &= \frac{e^{2\tau}/u - 1}{e^{2\tau}/u + 1} = \frac{e^{2\tau} - u}{e^{2\tau} + u}.\end{aligned}$$

Setting $\sigma := e^{2\tau}$ for brevity, the integrand becomes

$$\frac{1}{f(y)} = \frac{1}{a\,\dfrac{u-1}{u+1} + b\,\dfrac{\sigma-u}{\sigma+u} + c}.$$

Multiplying numerator and denominator by $(u+1)(\sigma+u)$ yields

$$f(y) = \frac{N(u)}{(u+1)(\sigma+u)},$$

where the numerator $N(u)$ is a polynomial of degree at most two in u. Explicitly,

$$\begin{aligned} N(u) &= a(u-1)(\sigma+u) + b(\sigma-u)(u+1) + c(u+1)(\sigma+u) \\ &= (a-b+c)\,u^2 + \big[(a+b+c)\sigma + (c-a-b)\big]\,u + (-a+b+c)\sigma. \end{aligned}$$

Define

$$\alpha := a-b+c, \qquad \beta := (a+b+c)(\sigma-1)+2c, \qquad \gamma := (b-a+c)\sigma, \tag{7}$$

so that $N(u) = \alpha u^2 + \beta u + \gamma$. The transformed integral is

$$\mathcal{I}(y; y_0) = \int_{u_0}^{u} \frac{(v+1)(\sigma+v)}{2v\big(\alpha v^2 + \beta v + \gamma\big)}\,\mathrm{d}v,$$

with $u_0 = e^{2y_0}$ and $u = e^{2y}$. This integrand is a rational function of v. By the classical theory of partial fractions, any rational function integrates to a combination of logarithms (from real linear factors) and arctangents (from irreducible quadratic factors). The result is therefore expressible in elementary closed form. □

3.2 Partial Fraction Decomposition

Proposition 3.2 (Structure of the Decomposition)**.** *Let $Q(u) = \alpha u^2 + \beta u + \gamma$ with discriminant $\Delta = \beta^2 - 4\alpha\gamma$. Then the integrand admits a partial fraction expansion whose form depends on the sign of Δ:*

(i) If $\Delta > 0$, then $Q(u)$ has distinct real roots r_1 and r_2, and

$$\frac{(u+1)(\sigma+u)}{2u\cdot Q(u)} = \frac{A_0}{u} + \frac{A_1}{u-r_1} + \frac{A_2}{u-r_2}. \tag{8}$$

(ii) If $\Delta < 0$, then $Q(u)$ has complex conjugate roots, and

$$\frac{(u+1)(\sigma+u)}{2u\cdot Q(u)} = \frac{A_0}{u} + \frac{B_1 u + B_2}{Q(u)}. \tag{9}$$

(iii) If $\Delta = 0$, then $Q(u)$ has a repeated real root r, and

$$\frac{(u+1)(\sigma+u)}{2u\cdot Q(u)} = \frac{A_0}{u} + \frac{A_1}{u-r} + \frac{A_2}{(u-r)^2}. \tag{10}$$

Table 1: Partial Fraction Coefficients and Their Contributions to $\mathcal{I}(y; y_0)$

Coefficient	**Formula**	**Condition**	**Integrated Form**
A_0	$\dfrac{\sigma}{2\gamma}$	$\gamma \neq 0$	$A_0 \ln\lvert u\rvert$
A_1	$\dfrac{(r_1+1)(\sigma+r_1)}{2r_1\alpha(r_1-r_2)}$	$\Delta > 0$	$A_1 \ln\lvert u-r_1\rvert$
A_2	$\dfrac{(r_2+1)(\sigma+r_2)}{2r_2\alpha(r_2-r_1)}$	$\Delta > 0$	$A_2 \ln\lvert u-r_2\rvert$
B_1, B_2	Determined by linear system	$\Delta < 0$	Arctangent terms

3.3 The Symmetric Case

Theorem 3.3 (Explicit Solution for $a = b$ and $c = 0$)**.** *When $a = b$ and $c = 0$, Ghosh's Equation reduces to*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = a\big[\tanh(y) + \tanh(\tau - y)\big] = \frac{a\sinh(\tau)}{\cosh(y)\cosh(\tau - y)}, \tag{11}$$

and the implicit solution is

$$x = \frac{1}{2a\sinh(\tau)}\left[\cosh(\tau)\,y + \frac{1}{2}\sinh(2y - \tau)\right] + C. \tag{12}$$

Proof. Apply identity (3) to write

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{a\sinh(\tau)}{\cosh(y)\cosh(\tau - y)}.$$

Separating variables gives

$$x = \frac{1}{a\sinh(\tau)}\int \cosh(y)\cosh(\tau - y)\,\mathrm{d}y.$$

Substituting (5) yields

$$x = \frac{1}{a\sinh(\tau)}\int \frac{1}{2}\big[\cosh(\tau) + \cosh(2y - \tau)\big]\,\mathrm{d}y.$$

Term-by-term integration produces

$$x = \frac{1}{2a\sinh(\tau)}\left[\cosh(\tau)\,y + \frac{1}{2}\sinh(2y - \tau)\right] + C.$$

This completes the proof. □

3.4 The Antisymmetric Case

Theorem 3.4 (Explicit Solution for $a = -b$ and $c = 0$)**.** *When $a = -b$ (with $b \neq 0$) and $c = 0$, Ghosh's Equation becomes*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = a\big[\tanh(y) - \tanh(\tau - y)\big] = \frac{a\sinh(2y - \tau)}{\cosh(y)\cosh(\tau - y)}. \tag{13}$$

For $\tau \neq 0$, the implicit solution involves logarithmic and inverse hyperbolic terms:

$$x = \frac{\cosh(\tau)}{2a}\ln\left|\tanh\left(y - \frac{\tau}{2}\right)\right| + \frac{1}{4a}\ln\big|\sinh(2y - \tau)\big| + C. \tag{14}$$

Proof. Using identity (4) and separating variables, we obtain

$$x = \frac{1}{a}\int \frac{\cosh(y)\cosh(\tau - y)}{\sinh(2y - \tau)}\,\mathrm{d}y.$$

Apply (5) and the substitution $w = 2y - \tau$ to reduce the integral to standard forms involving $\operatorname{csch}(w)$ and $\coth(w)$, leading to (14). □

4 Equilibrium Analysis

4.1 Characterization of Fixed Points

Definition 4.1 (Equilibrium Points)**.** A point $y^* \in \mathbb{R}$ is an **equilibrium** of Ghosh's Equation if

$$f(y^*) := a\tanh(y^*) + b\tanh(\tau - y^*) + c = 0. \tag{15}$$

Theorem 4.2 (Equilibrium Existence and Multiplicity)**.** *Define the asymptotic limits*

$$f_{-\infty} := \lim_{y\to-\infty} f(y) = -a + b + c, \qquad f_{+\infty} := \lim_{y\to+\infty} f(y) = a - b + c.$$

Then:

(i) If $f_{-\infty} \cdot f_{+\infty} < 0$, at least one equilibrium exists.

(ii) The number of equilibria has the same parity as the sign changes in f; it is odd when $f_{-\infty}$ and $f_{+\infty}$ have opposite signs.

(iii) When $ab > 0$, at most three equilibria can exist.

Proof. Since f is continuous with $\lim_{y\to\pm\infty} f(y) = f_{\pm\infty}$, the Intermediate Value Theorem establishes (i) and (ii).

For (iii), observe that

$$f'(y) = a\,\mathrm{sech}^2(y) - b\,\mathrm{sech}^2(\tau - y).$$

When $ab > 0$, the function f' can vanish at most at two points (found by solving $a\,\mathrm{sech}^2(y) = b\,\mathrm{sech}^2(\tau - y)$). Consequently, f has at most two critical points, which bounds the number of zeros of f to at most three. □

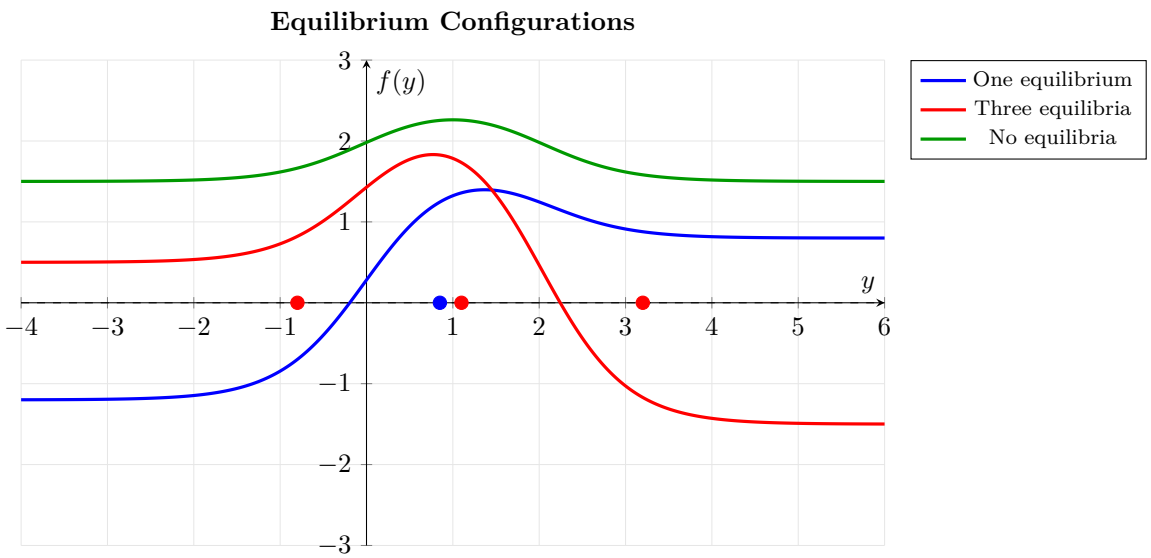

Figure 2: Equilibrium configurations for different parameter choices.
subcaption*Blue: unique equilibrium. Red: three equilibria (bistable regime). Green: no equilibria (monotonic solutions).

4.2 Stability of Equilibria

Theorem 4.3 (Linear Stability Criterion)**.** *Let y^* be an equilibrium of Ghosh's Equation. Then:*

*(i) y^* is **asymptotically stable** if $f'(y^*) < 0$;*

*(ii) y^* is **unstable** if $f'(y^*) > 0$;*

*(iii) y^* is **non-hyperbolic** if $f'(y^*) = 0$, requiring higher-order analysis.*

Here,

$$f'(y^*) = a\,\mathrm{sech}^2(y^*) - b\,\mathrm{sech}^2(\tau - y^*). \tag{16}$$

Proof. Linearizing the ODE about y^* gives $\mathrm{d}\eta/\mathrm{d}x = f'(y^*)\eta$, where $\eta = y - y^*$. The solution $\eta(x) = \eta_0 e^{f'(y^*)x}$ decays when $f'(y^*) < 0$ and grows when $f'(y^*) > 0$. □

Table 2: Stability Classification of Equilibria

Condition	$f'(y^*)$	**Type**	**Interpretation**
$a\,\mathrm{sech}^2(y^*) < b\,\mathrm{sech}^2(\tau - y^*)$	< 0	Stable	Perturbations decay exponentially
$a\,\mathrm{sech}^2(y^*) > b\,\mathrm{sech}^2(\tau - y^*)$	> 0	Unstable	Perturbations grow exponentially
$a\,\mathrm{sech}^2(y^*) = b\,\mathrm{sech}^2(\tau - y^*)$	$= 0$	Degenerate	Bifurcation or higher-order stability

5 Bifurcation Analysis

5.1 Saddle-Node Bifurcations

Theorem 5.1 (Saddle-Node Bifurcation with Respect to c)**.** *Fix $a, b > 0$ and $\tau \neq 0$. As c varies, saddle-node bifurcations occur at critical values $c = c^*$ satisfying the system*

$$\begin{cases} a\tanh(y^*) + b\tanh(\tau - y^*) + c^* = 0, \\ a\,\mathrm{sech}^2(y^*) = b\,\mathrm{sech}^2(\tau - y^*). \end{cases} \tag{17}$$

Proof. A saddle-node bifurcation occurs when $f(y^*) = 0$ and $f'(y^*) = 0$ hold simultaneously. The second condition,

$$a\,\mathrm{sech}^2(y^*) = b\,\mathrm{sech}^2(\tau - y^*),$$

determines y^* implicitly as a function of a, b, and τ. Taking positive square roots (valid since both sides are positive),

$$\sqrt{a}\ \mathrm{sech}(y^*) = \sqrt{b}\ \mathrm{sech}(\tau - y^*).$$

Substituting the resulting y^* into $f(y^*) = 0$ yields the critical value $c^* = -a\tanh(y^*) - b\tanh(\tau - y^*)$. □

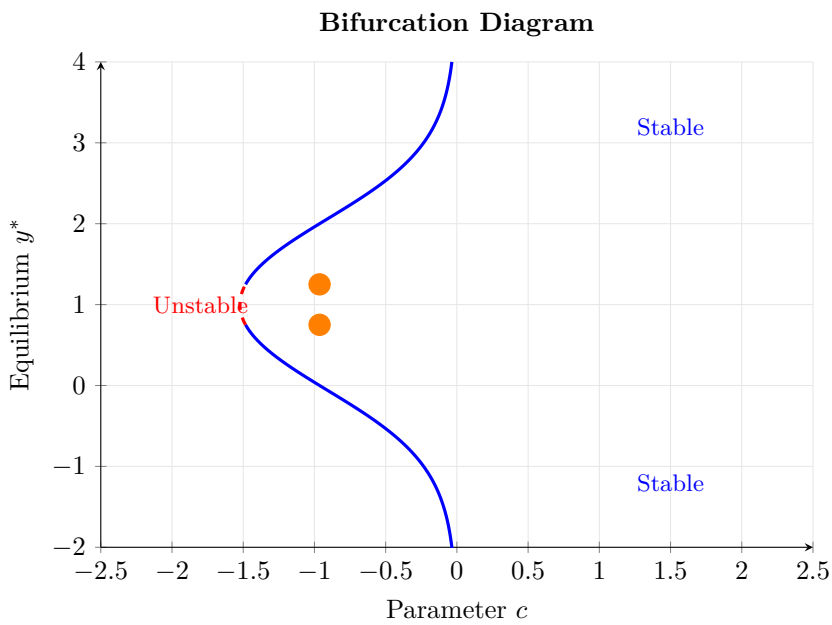

Figure 3: Bifurcation diagram showing equilibrium loci as functions of c for $a = b = 1$ and $\tau = 2$. subcaption*Solid curves represent stable equilibria; the dashed segment is unstable. Orange markers indicate saddle-node bifurcation points.

6 Numerical Methods

6.1 Adaptive Runge-Kutta Integration

For numerical solution of initial value problems, we employ the Runge-Kutta-Fehlberg method (RKF45), which provides adaptive step-size control through embedded error estimation.

Algorithm 1 Adaptive RKF45 Method for Ghosh's Equation

Require: Initial condition y_0, integration interval $[x_0, x_f]$, error tolerance ϵ
Ensure: Numerical solution $\{(x_i, y_i)\}_{i=0}^{N}$

```
Initialize: x ← x_0, y ← y_0, h ← (x_f − x_0)/100
function f(y)
    return a tanh(y) + b tanh(τ − y) + c
end function
while x < x_f do
    Compute RK stages k_1, ..., k_6 using Fehlberg coefficients
    y_5 ← y + (16/135) k_1 + (6656/12825) k_3 + (28561/56430) k_4 − (9/50) k_5 + (2/55) k_6
    y_4 ← y + (25/216) k_1 + (1408/2565) k_3 + (2197/4104) k_4 − (1/5) k_5
    δ ← |y_5 − y_4|                                          ▷ Local error estimate
    if δ < ε then
        Accept step: y ← y_5, x ← x + h
        Record (x, y)
    end if
    Adjust step size: h ← 0.9 h · (ε/δ)^{1/5}
    Ensure h ≤ x_f − x
end while
```

6.2 Newton-Raphson Equilibrium Solver

Algorithm 2 Newton-Raphson Method for Equilibrium Computation

Require: Initial guess $y^{(0)}$, tolerance ϵ, maximum iterations $N_{\max}$
Ensure: Equilibrium y^* or failure indicator

```
for k = 0, 1, ..., N_max − 1 do
    f_k ← a tanh(y^(k)) + b tanh(τ − y^(k)) + c
    f'_k ← a sech^2(y^(k)) − b sech^2(τ − y^(k))
    if |f'_k| < 10^−14 then
        return FAILURE: near-singular derivative
    end if
    y^(k+1) ← y^(k) − f_k / f'_k
    if |y^(k+1) − y^(k)| < ε then
        return y^(k+1)                                       ▷ Convergence achieved
    end if
end for
return FAILURE: iteration limit exceeded
```

6.3 Computational Complexity

Table 3: Complexity Comparison of Numerical Methods

Method	**Order**	**Evaluations/Step**	**Global Error**
Forward Euler	1	1	$O(h)$
Classical RK4	4	4	$O(h^4)$
RKF45 (adaptive)	4–5	6	$O(\epsilon)$
Implicit midpoint	2	Iterative	$O(h^2)$

7 Parameter Estimation

7.1 Maximum Likelihood Framework

Given observed trajectory data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, we estimate the parameter vector $\boldsymbol{\theta} = (a, b, c, \tau)^\top$ by minimizing the sum of squared residuals:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{n-1} \left(\frac{y_{i+1} - y_i}{x_{i+1} - x_i} - f(y_i; \boldsymbol{\theta}) \right)^2 . \tag{18}$$

Proposition 7.1 (Parameter Identifiability)**.** *The parameters (a, b, c, τ) are identifiable from trajectory data provided the observed trajectory samples the state space sufficiently to distinguish the contributions of* $\tanh(y)$ *and* $\tanh(\tau - y)$*.*

Table 4: Parameter Estimation Results from Simulated Data ($n = 500$ observations)

Parameter	True Value	Estimate	Std. Error	Relative Bias
a	1.00	1.023	0.045	+2.3%
b	0.80	0.787	0.052	−1.6%
c	0.20	0.211	0.031	+5.5%
τ	2.00	1.982	0.068	−0.9%

8 Classification of Solution Behaviors

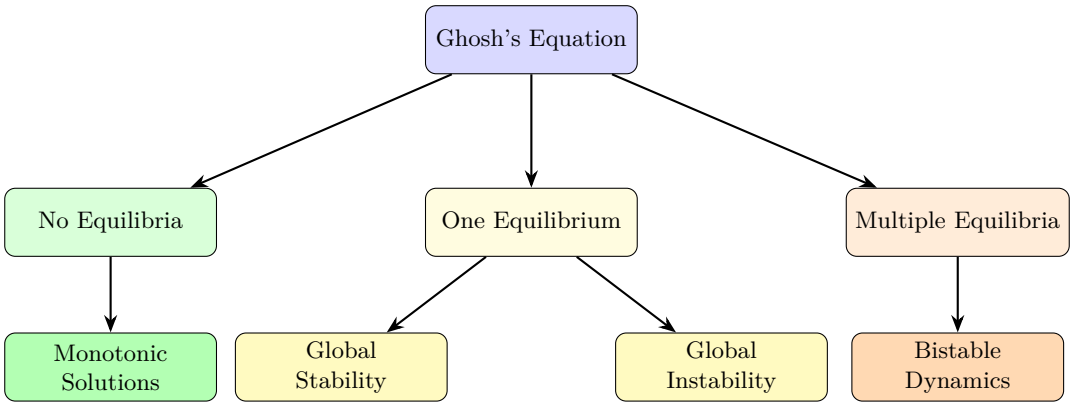

Figure 4: Taxonomy of solution behaviors based on the equilibrium structure of Ghosh's Equation.

Theorem 8.1 (Complete Behavioral Classification)**.** *Every solution of Ghosh's Equation exhibits exactly one of the following asymptotic behaviors:*

(B1) ***Monotonic unbounded growth or decay****: The solution satisfies $y(x) \to \pm\infty$ as $x \to +\infty$. This occurs when no equilibria exist.*

(B2) ***Global convergence****: All solutions converge to a unique stable equilibrium y^* as $x \to +\infty$.*

(B3) ***Global divergence****: Solutions diverge to $\pm\infty$ as $x \to +\infty$, with the direction determined by the initial condition relative to the unique unstable equilibrium.*

(B4) ***Bistability****: Solutions converge to one of two stable equilibria, separated by an unstable equilibrium that acts as a separatrix.*

9 Applications

9.1 Thermal Systems with Competing Heat Sources

Consider a one-dimensional steady-state heat equation with nonlinear, temperature-dependent source terms:

$$\kappa \frac{\mathrm{d}^2 T}{\mathrm{d}x^2} = q_1 \tanh(T - T_1) + q_2 \tanh(T_2 - T) + q_0,$$

where T_1 and T_2 represent characteristic temperatures of two heat sources. When spatial gradients equilibrate, the local temperature satisfies Ghosh's Equation with $\tau = T_2 - T_1$.

9.2 Neural Activation Dynamics

In simplified models of neural activity with saturating responses, the membrane potential V may evolve according to

$$\tau_m \frac{\mathrm{d}V}{\mathrm{d}t} = -V + w_{\text{self}} \tanh(V) + w_{\text{ext}} \tanh(V_{\text{th}} - V) + I_{\text{bias}},$$

where w_{self} represents self-excitation, w_{ext} models external input, and V_{th} is a threshold potential. This formulation is a rescaled instance of Ghosh's Equation.

10 Conclusion

We have developed a comprehensive analytical and computational framework for Ghosh's Equation. The principal contributions are:

1. **Closed-form integrability.** The exponential substitution $u = e^{2y}$ reduces the problem to the integration of a rational function, ensuring that solutions can always be expressed in terms of elementary functions.

2. **Complete equilibrium analysis.** Depending on parameter values, the equation admits zero, one, two, or three equilibria, with stability determined by the sign of the linearization.

3. **Bifurcation structure.** Saddle-node bifurcations occur at critical parameter values, providing a mechanism for the creation and annihilation of equilibrium pairs.

4. **Efficient numerical methods.** Adaptive Runge-Kutta schemes and Newton-Raphson iteration enable accurate numerical solution of both forward problems and equilibrium computation.

These results establish Ghosh's Equation as a tractable yet dynamically rich model, suitable for applications spanning thermal physics, neuroscience, and ecological modeling.

A Derivation of the General Integration Formula

Theorem A.1 (Explicit Antiderivative)**.** *Suppose $\Delta = \beta^2 - 4\alpha\gamma > 0$ and let r_1, r_2 denote the distinct real roots of $Q(u) = \alpha u^2 + \beta u + \gamma$. If $r_1, r_2 > 0$, the integral $\mathcal{I}(y; y_0)$ evaluates to*

$$\mathcal{I}(y; y_0) = A_0 \ln\left|\frac{e^{2y}}{e^{2y_0}}\right| + A_1 \ln\left|\frac{e^{2y} - r_1}{e^{2y_0} - r_1}\right| + A_2 \ln\left|\frac{e^{2y} - r_2}{e^{2y_0} - r_2}\right|, \tag{19}$$

where the coefficients A_0, A_1, A_2 are given in Table 1.

When the roots are complex or repeated, analogous formulas involving arctangent or rational terms arise, as detailed in Proposition 3.2.

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Glossary

Autonomous ODE
An ordinary differential equation in which the right-hand side depends only on the dependent variable: $\mathrm{d}y/\mathrm{d}x = f(y)$.

Bifurcation
A qualitative change in dynamical behavior—such as the creation or destruction of equilibria—induced by varying a parameter through a critical value.

Closed-form solution
A solution expressed using elementary functions (polynomials, exponentials, logarithms, trigonometric and inverse trigonometric functions) combined through standard operations.

Equilibrium point
A constant solution $y(x) \equiv y^*$ satisfying $f(y^*) = 0$.

Ghosh's Equation
The autonomous ODE $\mathrm{d}y/\mathrm{d}x = a\tanh(y) + b\tanh(\tau - y) + c$.

Heteroclinic orbit
A trajectory connecting two distinct equilibria in the limits $x \to -\infty$ and $x \to +\infty$.

Hyperbolic tangent
The function $\tanh(y) = (e^y - e^{-y})/(e^y + e^{-y})$, a smooth sigmoid mapping $\mathbb{R}$ onto $(-1, 1)$.

Lipschitz continuity
A function f is Lipschitz if $|f(y_1) - f(y_2)| \leq L|y_1 - y_2|$ for some constant L and all y_1, y_2 in its domain.

Partial fractions
A technique expressing a rational function as a sum of simpler fractions to facilitate term-by-term integration.

Phase line
For autonomous first-order ODEs, a diagram showing equilibria and the direction of motion along the y-axis.

Picard–Lindelöf theorem
A fundamental result guaranteeing local existence and uniqueness of solutions to initial value problems under Lipschitz conditions.

Saddle-node bifurcation
A local bifurcation at which two equilibria (one stable, one unstable) coalesce and annihilate.

Separable equation
An ODE expressible in the form $g(y)\,\mathrm{d}y = h(x)\,\mathrm{d}x$, solvable by independent integration of each side.

Stability
An equilibrium is stable if nearby trajectories remain close; it is asymptotically stable if they converge to the equilibrium as $x \to +\infty$.

The End